

{ Mendelian Randomization (MR)

MR uses genetic variants as IVs to estimate the causal effect of an exposure (Ex. BMI) on an outcome (Ex. Disease Risk)

★ Mendel's Laws of Inheritance

i) Law of Segregation (分離律)

Alleles separate at meiosis and a randomly selected allele is transmitted to offspring

《Punnett Square》

		pollen ♂		
		B	b	
pistil ♀	B	BB	Bb	B: dominant allele (顯性) b: recessive allele (隱性)
	b	Bb	bb	

The genotypic ratio $BB:Bb:bb = 1:2:1$

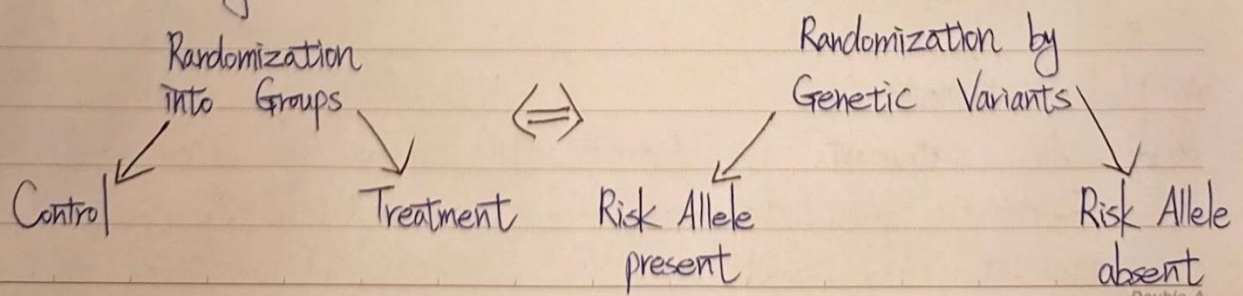
Since $BB, Bb \rightarrow$ purple
 $bb \rightarrow$ white

The phenotypic ratio purple:white = 3:1

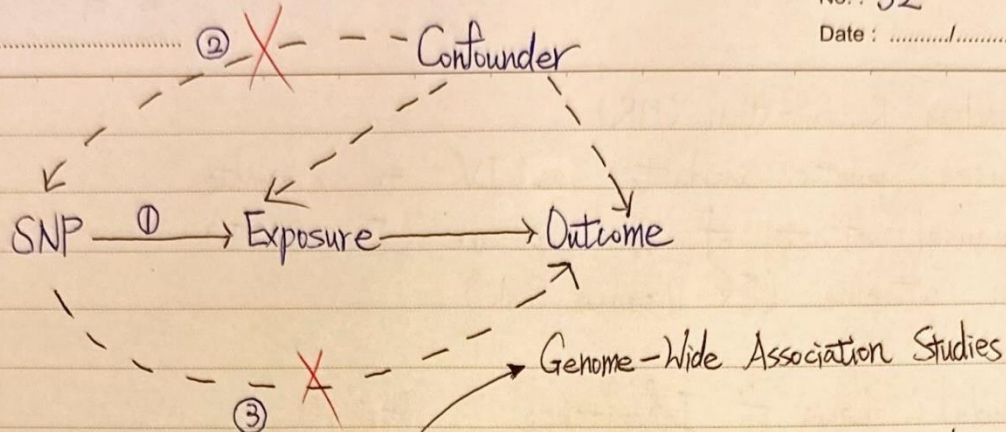
ii) Law of Independent Assortment (獨立分配律)

Alleles for separate traits are transmitted independently of one another.

★ MR uses genetic variation to mimic RCT



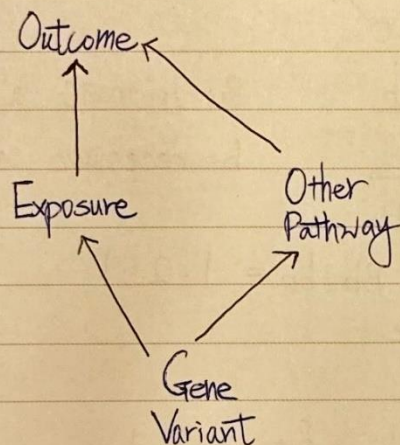
Subject :



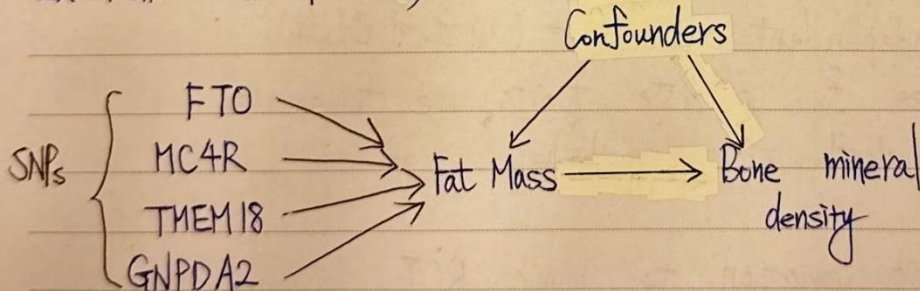
- ① ✓ ⇒ Modern GWAS have identified many causal variants.
 ② ✓ ⇒ Almost Comes for free due to Mendel's Law
 ③ ? ⇒ Problematic because of horizontal pleiotropy.

基因多效性

It is a condition in which a single gene or (Violation) genetic variant influences multiple phenotypic traits.



Ex. Palmer et al (2012) Stat Method Res



* Weak instruments

An instrument is (statistically) weak if the F-statistic associated to the observed Z-A association is small. (< 10)

Def (Econometric)

(ver 1) Instrument variables that have a low correlation w/ endogenous regressors.

(ver 2) Instruments that explain little of the variation in endogenous regressors.

★ Three serious problems arise when instrument is weak

- i) Yield effect estimate w/ wide CI.
- ii) Amplify bias due to the violations of ② and ③
- iii) Even with a valid instrument and large sample size, weak instruments introduce bias in the standard IV estimator.

★ The importance of strong instrument
define a strong instrument Z high correlated w/ endogenous variable A

$$Y_i = \beta_0 + \beta_1 A_i + \epsilon_i$$

the IV estimator of β_1 is

$$\hat{\beta}_1 = \frac{\sum Y_i (Z_i - \bar{Z})}{\sum (A_i - \bar{A})(Z_i - \bar{Z})} = \frac{S_{YZ}}{S_{AZ}}$$

$$\text{Var}(\hat{\beta}_1) = \frac{\sum (Z_i - \bar{Z})^2 \sigma^2}{S_{AZ}^2}$$

$$\hat{\sigma}_{IV}^2 = \frac{\sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 A_i)^2}{n-2} \rightarrow \hat{\sigma}^2$$

$$\widehat{\text{Var}}(\hat{\beta}_1) = \frac{\hat{\sigma}_{IV}^2 \sum (Z_i - \bar{Z})^2}{S_{AZ}^2} = \frac{\hat{\sigma}_{IV}^2}{\left(\frac{S_{AZ}^2}{S_{ZZ} S_{AA}} \right) S_{AA}}$$

$$= \frac{\hat{\sigma}_{IV}^2}{r_{AZ}^2 S_{AA}}$$

$$\frac{\widehat{\text{Var}}(\hat{\beta}_1)}{\widehat{\text{Var}}(\check{\beta}_1)} = \frac{\frac{\hat{\Delta}_{IV}^2}{r_{ZA}^2 SAA}}{\frac{\hat{\sigma}^2}{SAA}} \approx \frac{1}{r_{ZA}^2} \quad \left(\begin{array}{l} \text{In large sample} \\ \frac{\hat{\Delta}_{IV}^2}{\hat{\sigma}^2} \approx 1 \end{array} \right)$$

OLS

the $r_{AZ}^2 < 1$, $\widehat{\text{Var}}(\hat{\beta}_1) > \widehat{\text{Var}}(\check{\beta}_1)$

If $r_{AZ} = 0.5$, $\frac{\text{se}(\hat{\beta}_1)}{\text{se}(\check{\beta}_1)} \approx 2$

$= 0.1 \quad \approx 10$

recall that in large samples a 95% CI $\approx$ estimate ± 2 Se
suppose $\check{\beta}_1 \approx \hat{\beta}_1 = 5$ and $\text{Se}(\check{\beta}_1) = 1$

then 95% CI of $\check{\beta}_1$ is $5 \pm 2 \times 1 \Leftrightarrow [3, 7]$

If $r_{AZ} = 0.5$, 95% CI of $\hat{\beta}_1$ is $5 \pm 2 \times 2 \Leftrightarrow [1, 9]$

$= 0.1$, " $5 \pm 2 \times 10 \Leftrightarrow [-15, 25]$

Thus, If you do have to use IV estimation, then you must search for a strong instrument, one that is highly correlated w/ endogenous A.

★ Hausman Test for Endogeneity

• Rationale

$$Y = \beta_0 + \beta_1 A + \epsilon$$

$$A = \alpha_0 + \gamma_1 Z + v = E(A|Z) + v = \hat{A} + \hat{v}$$

then

$$Y = \beta_0 + \beta_1 (\hat{A} + \hat{v}) + \epsilon$$

$$= \beta_0 + \beta_1 \hat{A} + \beta_1 \hat{v} + \epsilon$$

$$\triangleq \beta_0 + \beta_1 \hat{A} + \delta \hat{v} + \epsilon \quad (H_0: \beta_1 = \delta)$$

$$= \beta_0 + \beta_1 (\hat{A} + \hat{v}) + (\delta - \beta_1) \hat{v} + \epsilon$$

$$= \beta_0 + \beta_1 A + \delta \hat{v} + \epsilon \quad (H_0: \delta = 0)$$

Consider $Y = \beta_0 + \beta_A A + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$

$\checkmark$ Z_1, Z_2 instruments

Step 1. Estimate 1st stage

$$A = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + \gamma_1 Z_1 + \gamma_2 Z_2 + v \Rightarrow \text{get } \hat{v}$$

(If more than one explanatory variables being tested for endogeneity, repeat this for each)

Step 2. Include residuals $\hat{v}$ as an explanatory variable

$$Y = \beta_0 + \beta_A A + \delta \hat{v} + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

Estimate it by OLS and employ the usual T-test

$$H_0: \delta = 0 \quad \Leftrightarrow \quad \text{Cov}(A, \varepsilon) = 0$$

$$H_1: \delta \neq 0 \quad \Leftrightarrow \quad \text{Cov}(A, \varepsilon) \neq 0$$

(If more than one variable is being test, the test will be on F-test of joint significance of the coeffs on the included residuals)

★ General form

$$y = AX + X\theta + \varepsilon \quad \checkmark \text{ instruments } Z$$

by FGLS thm, for OLS

$$\hat{\theta} = (A^T(I_n - P_X)A)^{-1} A^T(I_n - P_X)y$$

and for 2SLS

$$\hat{\theta} = (A^T(P_Z - P_X)A)^{-1} A^T(P_Z - P_X)y$$

then

$$V(\hat{\theta}) = \sigma^2 (A^T(I_n - P_X)A)^{-1}$$

$$V(\hat{\theta}) = \sigma^2 (A^T(P_Z - P_X)A)^{-1}$$

$$\text{let } d = \hat{\theta} - \theta$$

$$V(d) = V(\hat{\theta}) + V(\theta) - 2 \text{Cov}(\hat{\theta}, \theta)$$

$$\text{under } H_0, \text{Cov}(\hat{\theta}, \theta) = V(\theta)$$

$$V(d) = V(\hat{\xi}) - V(\check{\xi})$$

Inserting the results, we have

$$H = d^T [V(\hat{\xi}) - V(\check{\xi})] d$$

$$= \frac{d^T [(A^T(P_Z - P_X)A)^{-1} - (A^T(I_n - P_X)A)^{-1}] d}{\Delta^2}$$

★ Over-identification Test

When

$$\text{No. of IVs} \xrightarrow{m} > K \xleftarrow{\text{No. of endogenous regressors}}$$

- The model is overidentified meaning that there are more moments than free parameters.
- This is a restriction and it testable.

For a general over-identification test the null and alternative hypotheses are

$$H_0: E(Z^T \varepsilon) = 0 \quad \text{vs} \quad H_1: E(Z^T \varepsilon) \neq 0$$

- If the test indicates you should reject the null, then this is pretty clear evidence your model is mis-specified.
- Whenever the over-identification test implies rejection of the null, this usually means at least one of instruments would have a significant effect in structural equation.

Recall that standard IV estimator is special case of GMM estimator.

Generalized method of moment

Since $\begin{cases} Z_1, \dots, Z_m \\ X_1, \dots, X_n \end{cases}$ are exogenous

The instruments give us a set of moments

$$E(W^T \varepsilon) = 0 \Rightarrow E(W^T (y - D\beta)) = 0$$

Each of moment correspond to sample moment

$$g_n(\beta) = \frac{1}{n} W^T (y - D\beta) = \frac{1}{n} W^T \varepsilon$$

The intuition behind GMM is to choose estimator of β , $\hat{\beta}$ sets these sample moments as close to zero as possible.

Consider an weighting matrix M is p.s.d. Construct a quadratic form in the moment conditions. This gives GMM objective J_n .

$$J(\beta) = n g_n(\beta)^T M g_n(\beta)$$

A GMM estimator for β is the $\hat{\beta}$ that $\min_{\beta} J(\beta)$

$$\frac{dJ(\beta)}{d\beta} = 0$$

yields the GMM estimator

$$\hat{\beta} = (D^T M M^T D)^{-1} D^T M M^T y$$

• What is the optimal choice of M ?

$$S = \frac{1}{n} E([W^T \varepsilon][W^T \varepsilon]^T)$$

$$= \frac{1}{n} E(W^T \varepsilon \varepsilon^T W)$$

$$= \frac{1}{n} E(W^T \Omega W) \xrightarrow{\text{homoskedasticity}} \frac{1}{n} E^T E(W^T W)$$

Covariance matrix of moment condition g

homoskedasticity

We find that it reduces to IV estimator.

If we now set $\hat{M} = \hat{S}^{-1} = \left(\hat{\Sigma}^2 \frac{1}{n} \bar{M}^T \bar{M} \right)^{-1}$

We obtain Sargan test

$$\frac{\hat{\Sigma} \bar{M} (\bar{M}^T \bar{M})^{-1} \bar{M}^T \hat{\Sigma}}{\hat{\Sigma}^2} = n \left(\frac{\hat{\Sigma}^T \hat{\Sigma}}{\hat{\Sigma}^T \hat{\Sigma}} \right)$$

$$= n R^2 \sim \chi^2_{(m-k)}$$

equivalent

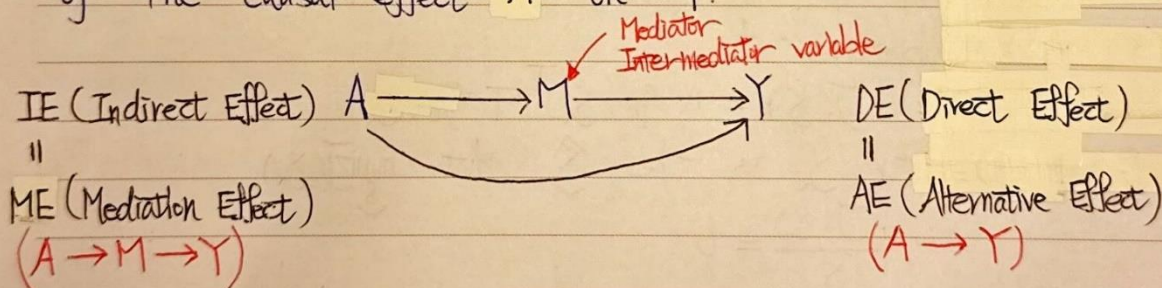
Regress $\hat{\Sigma}$ against $\begin{cases} Z_0, \dots, Z_m & \text{(excluded)} \\ X_1, \dots, X_r & \text{(included)} \end{cases} \Rightarrow R^2$

$$= (m-k) F \sim \chi^2_{(m-k)}$$

Compute F statistic ($H_0: \alpha_1 = \alpha_2 = \dots = \alpha_m = 0$)

Mediation

Mediation analysis address questions about mechanism M of the causal effect A on Y.



★ Prescriptive Interpretation (Policy making)

- It is the effect of A after prescribing or intervening on the mediator M.
- We may be interested in understanding what happen if a new policy is promoted and implemented.

★ Descriptive Interpretation (Mechanism investigation)

- It aim to investigate the natural mechanism that exposure affect outcome through mediator.